



Fig. 9: Author's prototype

This helps reduce errors by enabling accurate detection of logic levels. The receiver then reads 8 successive bits to reconstruct the transmitted byte. These bits are shifted into their appropriate positions to retrieve the complete byte. Synchronisation is maintained using the same `BITDELAY` interval (in milliseconds) between bits.

The reconstructed byte (ASCII value) is then converted into its corresponding character and displayed on the receiver's serial monitor. Padding each character with start and stop bits ('0' and '1') ensures distinct boundaries between bytes, allowing the receiver to recognise individual units reliably. The value of `BITDELAY` must be chosen carefully—long enough to detect level changes accurately, yet short enough to allow practical communication rates.

Details for our implemented setup. An optimum `BITDELAY`

value of 3 milliseconds is used, resulting in a data rate of:

$1/3\text{ms} = 1/0.003\text{s} = 333.33\text{bps}$ (bits per second).

Each character transmitted consists of 10 bits (start bit '0' + 8-bit ASCII + stop bit '1').

Using a `BITDELAY` of 1 millisecond would yield a rate of 1 kbps, but it causes bit skipping and erroneous communication. A 10-millisecond `BITDELAY` would result in 100bps, which is too slow for effective use.

Software

Two separate devices are used for wireless optical transmission, requiring two distinct codebases: one for the transmitter and one for the receiver. In the transmitter code, define the output pin and set the `BITDELAY`.

The receiver code must use the same `BITDELAY` to maintain

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synchronisation and avoid data loss. Select the appropriate board and upload the respective code.

Construction and testing

Begin by uploading the source code to both boards. Then, assemble the two modules as illustrated in the circuit diagram (see Fig. 3 and Fig. 5). It is advisable to consult the authors' prototype before starting construction.

Both modules are powered via USB. Ensure that the laser and photodiode modules are directly aligned with a clear line of sight.

The communication distance is not critical, provided the laser beam directly strikes the photodiode. Open the Serial Monitor, send a message from the transmitter, and observe the receiver capturing and displaying it using Li-Fi transmission. **EFY**

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